ethical concerns, and the need for robust policy frameworks to ensure safe and sustainable deployment.

The entire book spreads over *nine chapters*, describing the various generations of the prevailing drone technology and what is expected in the future. Each chapter ends with a summarization of the content and some review questions so that the reader can assess their understanding.

Chapter 1, i.e., Overview of Drone Technology, is the introductory chapter presenting a panoramic view of UAV evolution, highlighting their origins in early 20th-century military research and tracing their expansion into civilian, industrial, and research domains. It explains the core principles of drone operation – covering aerodynamics, propulsion, communication, and navigation – along with emerging technologies such as AI, GPS precision, and digital twin integration. The chapter classifies drones into generational stages, setting the conceptual foundation for the chapters that follow.

Chapter 2 introduces the first-generation drones, revisiting the formative era of drone development (1910s–1930s), characterized by manual control, simple fixed-wing structures, and rudimentary radio systems. It introduces pioneering innovations such as the *Aerial Target* (1916), *Kettering Bug* (1918), and *Queen Bee* (1935), emphasizing their contributions to early defense training and reconnaissance. The discussion also covers the foundational technologies – radio communication, mechanical servos, and piston propulsion – that shaped the first UAV architectures.

Chapter 3 moves into second-generation drones. The focus here shifts to the integration of remote control and GPS technologies during the 1980s–1990s. This generation introduced semi-automated navigation, improved endurance, and digital telemetry. The chapter explores how enhanced flight controllers, onboard sensors, and composite materials expanded UAV